THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · NCERT-TYPE CHAPTER TEST SET 1

Chapter 12: **Surface Areas and Volumes** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A toy is a cone of base radius 7 cm and height 24 cm mounted on a hemisphere of the same radius (the flat faces coincide). The total surface area of the toy is: 1

(a) 836 cm² **(b)** 550 cm² **(c)** 858 cm² **(d)** 1012 cm²

2. A capsule is a cylinder of radius 3.5 cm and length 10 cm (straight part) with a hemisphere of the same radius fixed at each end. The total surface area of the capsule is: 1

(a) 297 cm² **(b)** 748 cm² **(c)** 451 cm² **(d)** 374 cm²

3. A solid is a cube of edge 14 cm surmounted by a hemisphere of diameter 14 cm (the hemisphere rests on a face of the cube). The total surface area of the solid is: 1

(a) 1022 cm² **(b)** 1176 cm² **(c)** 1330 cm² **(d)** 1484 cm²

4. From a solid cylinder of height 24 cm and base radius 7 cm, a conical cavity of the same height and the same base radius is hollowed out. The total surface area of the remaining solid is: 1

(a) 1760 cm² **(b)** 1364 cm² **(c)** 1606 cm² **(d)** 1914 cm²

5. A wooden article is made by scooping out a hemisphere of radius 3.5 cm from each end of a solid cylinder of height 10 cm and base radius 3.5 cm. The total surface area of the article is: 1

(a) 451 cm² **(b)** 297 cm² **(c)** 220 cm² **(d)** 374 cm²

6. A solid metallic cylinder of base radius 6 cm and height 32 cm is melted and recast into solid spheres of radius 3 cm. The number of spheres obtained is: 1

(a) 38 **(b)** 64 **(c)** 32 **(d)** 30

7. A solid metallic cone of base radius 6 cm and height 24 cm is melted and recast into a solid sphere. The radius of the sphere is: 1

(a) 6 cm **(b)** 7 cm **(c)** 5 cm **(d)** 12 cm

8. A toy is a cone of base radius 21 cm and height 28 cm mounted on a hemisphere of the same radius (the flat faces coincide). The total surface area of the toy is: 1

(a) 4620 cm² **(b)** 2310 cm² **(c)** 5082 cm² **(d)** 6468 cm²

9. A capsule is a cylinder of radius 7 cm and length 14 cm (straight part) with a hemisphere of the same radius fixed at each end. The total surface area of the capsule is: 1

(a) 924 cm² **(b)** 1540 cm² **(c)** 2464 cm² **(d)** 1232 cm²

10. A solid is a cube of edge 28 cm surmounted by a hemisphere of diameter 28 cm (the hemisphere rests on a face of the cube). The total surface area of the solid is: 1

(a) 5936 cm² **(b)** 5320 cm² **(c)** 4704 cm² **(d)** 4088 cm²

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A solid metallic sphere of radius 6 cm is melted and drawn into a wire of circular cross-section of radius 2 cm. Find the length of the wire. 2

12. A solid metal cuboid 24 cm by 12 cm by 9 cm is melted and recast into cubes of edge 6 cm. Find the number of cubes obtained. 2

13. A solid sphere of radius 3 cm is completely dropped into water in a cylindrical vessel of base radius 3 cm, and no water spills. Find the rise in the water level. 2

14. A solid metallic sphere of radius 6 cm is melted and drawn into a wire of circular cross-section of radius 3 cm. Find the length of the wire. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A toy is a cone of base radius 21 cm and height 10 cm mounted on a hemisphere of the same radius. Find the volume of the toy. 3

16. A capsule is a cylinder of radius 10.5 cm and length 10 cm (straight part) with a hemisphere of the same radius fixed at each end. Find the volume of the capsule. 3

17. From a solid cylinder of height 12 cm and base radius 7 cm, a conical cavity of the same height and base radius is hollowed out. Find the volume of the remaining solid. 3

OR

A toy is a cone of base radius 21 cm and height 20 cm mounted on a hemisphere of the same radius. Find the volume of the toy. 3

18. A capsule is a cylinder of radius 10.5 cm and length 20 cm (straight part) with a hemisphere of the same radius fixed at each end. Find the volume of the capsule. 3

OR

From a solid cylinder of height 24 cm and base radius 7 cm, a conical cavity of the same height and base radius is hollowed out. Find the volume of the remaining solid. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Water from a full cylindrical tank of radius 7 cm and height 10 cm is poured into an empty rectangular tank of base 11 cm by 7 cm. Find the height of water in the rectangular tank. 5

OR

A conical vessel of radius 6 cm and height 8 cm is full of water. The water is poured into a cylindrical vessel of base radius 4 cm. Find the height of the water in the cylindrical vessel. 5

20. Water from a full cylindrical tank of radius 7 cm and height 12 cm is poured into an empty rectangular tank of base 22 cm by 14 cm. Find the height of water in the rectangular tank. 5

OR

A conical vessel of radius 6 cm and height 12 cm is full of water. The water is poured into a cylindrical vessel of base radius 3 cm. Find the height of the water in the cylindrical vessel. 5

END OF QUESTION PAPER — Total: 40 Marks